

Game Arcade — UI build spec

A hypercasual games aggregator: forty games, one wallet, one social graph. This document is everything a designer or front-end engineer needs to build the interface — screens, chrome, components, states, game mechanics, copy and instrumentation. It describes the decided product; anything still open is listed in §11.

Reference build: *Game Arcade.dc.html* — a working prototype with STACK, REFLEX and DODGE playable and all fourteen screens navigable. Where this spec and the prototype disagree, this document wins; the prototype is tighter than production should be in two places, both flagged.

1. Product rules the UI has to express

Every one of these is a decision, not a suggestion. If a screen makes one of them hard to see, the screen is wrong.

RULE	WHAT IT MEANS ON SCREEN
Volume is the pitch	Forty games at launch, all free, nothing locked. Game count is visible on the shelf, in Browse and on the profile.
One wallet	Coins are global. They appear in the top bar on every hub screen and are never labelled per-game.
One social graph	Scores, duels and rivalries cut across all forty games. A friend's name can appear on any game's surface.
Sessions never end on purpose	A finished run auto-loads the next ranked game after three seconds, with a visible ring the player can tap to cancel.
Rivalry first	The ranked shelf and the auto-queue both prefer a game where a friend just beat you. Cold start substitutes a real, skill-matched global rival.
Twelve pinned, twenty-eight in Browse	The shelf is a curated twelve. Long-press pins and unpins. A thirteenth pin is refused until one is dropped.
Coins buy continues, never access	Retries and power-ups only. The shop says so in plain words.

2. Canvas, grid and spacing

Design canvas is 412 × 892 (Android, 1× density-independent pixels). The layout is a full-bleed single column — no cards, no page gutters, no rounded corners anywhere. Structure is carried entirely by rules and alignment.

TOKEN	VALUE	USE
Horizontal inset	14px	Every text edge on every screen. Content that bleeds (poster, banners) still insets its type by 14px.
Section rule	2px --color-divider	Between major sections and above the tab bar.
Row rule	1px --color-divider	Between list rows of the same kind.
Grid seam	2px divider as gap	Tile grids are a divider-coloured container with 2px gaps; tiles are background-coloured. No borders per tile.
Row padding	11–13px vertical	List rows. Tap targets are the whole row, never the label.
Tile min-height	62px (2-up) / 74px (3-up)	Shelf and picker grids.

Production correction: the prototype's tab-bar items and some list rows land near 40px tall. Build every interactive row and tab at **48px minimum**, growing the row padding rather than the type.

3. Type and colour

Archivo throughout — 800 weight for anything structural, 400/600 for prose. Numbers are always font-variant-numeric: tabular-nums, because scores sit in columns and must not jitter as they tick.

ROLE	SIZE / WEIGHT	WHERE
Run score	72px / 800, -0.04em	Results banner only. One per screen.
Display	40–46px / 800, -0.035em	Poster drop, onboarding headline, wallet balance.
Screen title	26–32px / 800, -0.03em	Duels, Inbox, profile handle, bracket round.
Stat figure	20–27px / 800	Score columns, stat tiles, feed post figures.
Row title	15–19px / 800	Game names, friend names, duel rows.
Body	13px / 400, 1.5	Explanatory lines, notification text.
Meta	11px / 400	Sub-lines: "Best 47 · 26 runs".
Kicker	10px / 400–600, 0.10–0.14em, uppercase	Section labels, banner kickers, column heads.

Production correction: the prototype drops kickers to 9px. Floor is 10px, and only for tracked uppercase labels — never for anything a player has to read to make a decision.

COLOUR	TOKEN	RULE
Ground	--color-bg	Every screen. Never a gradient, never a second background colour.
Ink	--color-text	All type, all game geometry, the "run over" results banner.
Accent	--color-accent	Reserved for four things: the daily drop poster, a personal-best banner, the live rivalry number, and the primary action on screen. Nothing else.
Accent tint	--color-accent-100	Row background meaning "this involves a rival or is urgent".
Accent deep	--color-accent-700	Accent-coloured type at 11px and below — the flat accent is not readable at that size.

4. Global chrome

Three variants, mutually exclusive. Hub screens get the status bar plus the tab bar; pushed screens get a back bar and keep the tab bar; a match has neither.

ELEMENT	CONTENTS AND BEHAVIOUR
Top bar hub screens	Left: wordmark (taps home). Then coin balance with a 9px accent square (taps Shop). Then STREAK + day count. Then INBOX with a 7px accent dot when unread. Separated by 1px vertical rules, closed by a 2px rule. Height 40px.
Back bar detail, drop, bracket, friend, inbox	"← BACK" flush left, screen label flush right in 13px/800 neutral. Back returns to the screen that pushed it, not a fixed parent.
Tab bar	Five equal columns: PLAY · FEED · DUELS · SHOP · ME. Labels flush left in their column at 10px/800, 0.06em. Active tab is accent type plus a 3px accent rule sitting on top of the 2px divider. DUELS carries a 6px accent dot when a ghost is waiting. PLAY stays active for Browse and Game detail.
Match chrome	Replaces all of the above: score pane left, rival ghost pane right on accent tint, 44px X to quit, and a 4px ghost-progress bar beneath. Nothing else — no tabs, no coin balance.

5. Screens

Fourteen screens. Each entry: what it is for, how it is built top to bottom, and what it does.

Build order should follow the numbers — 02, 05 and 06 are the product; everything else supports them.

01 · Onboarding — "Pick your three"

One screen, no account, no carousel. Kicker "STEP 1 OF 1 · 40 GAMES INSIDE", 38px headline **PICK YOUR THREE.**, one line of explanation, then a 3-up grid of twelve games (name, family, friends-on count). Tapping a tile inverts it to accent. The primary button is disabled-grey reading "PICK n MORE" until three are chosen, then accent "BUILD MY SHELF →". An underlined "Skip — just put me in a game" drops straight into STACK.

The three picks seed the twelve pinned slots; the other nine are filled by popularity. Never block the skip.

02 · Home — the ranked shelf

Four stacked bands:

a. Drop poster. Full-bleed accent. Kicker "TODAY'S VARIANT" with a right-aligned countdown, 42px two-line variant name, one line of rule, a white "PLAY NOW" button and a secondary line ("8 friends played / Top: MEHA 31") that pushes the Daily drop screen.

b. Ranked for you — three editorial rows. 54px ink thumbnail with the game's three-letter code (taps straight into a run), name at 17px/800, a neutral meta line, and a *reason line in accent* ("RAVI BEAT YOU BY 3", "9 FRIENDS PLAY THIS · YOU NEVER HAVE"). Right column: the player's best at 25px with its unit beneath.

c. Your shelf — the remaining nine pinned games as a 2-up grid: name, family label right-aligned, and best-or-"NEVER PLAYED" at the bottom. No reason lines here; ranking order is the only signal.

d. Footer — outlined "BROWSE 28 MORE →" plus the standing instruction "Long-press any game to pin or unpin".

Ranking: rival-ahead games first, then never-played, then the rest; ties broken by friends-on count. Reasons show on the top three only.

03 · Browse — all forty

Search field (2px ink border, square) with the live count right-aligned, then a horizontally scrolling family filter: ALL · TAP · SWIPE · TIMING · NUMBERS · MEMORY, selected chip inverted to ink. Below, a numbered index: NO · GAME · ON · BEST column heads, then one 10px-padded row per game — index, name with a "PINNED" flag, a sub-line ("SWIPE · RAVI ahead" / "· 26 runs" / "· never played"), friends-on count, and best right-aligned in a 54px column. Pinned rows sit on accent tint.

Tap opens the game's detail. Long-press (420ms) toggles the pin and shows an ink toast strip under the filters: "STACK PINNED TO YOUR SHELF" or, at twelve, "SHELF FULL — UNPIN ONE FIRST". Suppress the tap that follows a long-press.

04 · Game detail

96px ink code block beside the title block (family · plays today, 30px name, one-line rule description). Then a three-cell stat strip: YOUR BEST · FRIEND RANK · RUNS. Then the **ghost duel band** on accent tint — rival name and score at 19px, "Recorded 2h ago. Their ghost races you live.", and an accent "RACE IT" button. Then a leaderboard with FRIENDS / GLOBAL / THIS WEEK tabs and a season label; the player's own row is tinted. Then a daily-quest row with its coin value in accent. Full-width ink "PLAY" button closes the screen.

For the 37 non-playable games the button reads "PLAY NAME (SIMULATED)" in the prototype only; production drops the qualifier.

05 · In-match

Two-pane header: your live score left (26px), the rival's ghost right on accent tint, its number ticking up in accent as their recorded run replays against the clock. A 4px bar under the header fills toward the ghost's final score. The playfield takes the rest; the tab bar is gone. DODGE adds a two-button lane control strip at the bottom (56px cells, "← LEFT" / "RIGHT →"); STACK and REFLEX take a tap anywhere in the field. Arrow keys and space work for desk testing.

Blackout variant: the field background flips to `--color-neutral-900` on every third block. Geometry stays ink-on-ground; the falling piece is the only accent element in the field.

06 • Results — the retention screen

- a. Banner.** Accent if it is a personal best, ink otherwise. Kicker, the score at 72px, and one plain-language headline: "Up 4 on your old best. Posted to the feed." / "You took RAVI's ghost." / "RAVI's ghost held. 3 short."
- b. Three-cell strip.** YOUR BEST · rival ghost score (accent if beaten) · coins earned in accent.
- c. Near-miss band** (only when the run fell within 18% of the player's best without beating it). Accent tint, kicker "SO CLOSE", 22px "YOU WERE 4 BLOCKS OFF YOUR BEST", accent button "RETRY FREE" or "RETRY · 50 COINS", and a note stating retries left and that the retry restarts at the score reached.
- d. Auto-queue row.** A 46px conic-gradient ring counting three seconds down with the digit inside, "NEXT UP · AUTO-LOADING", the next game at 19px, its accent reason line, and "TAP TO CANCEL" right-aligned. Tapping anywhere on the row cancels and replaces it with "Queue stopped. DODGE is still waiting on your shelf." plus an outlined play button.
- e. Feed receipt** — one line stating exactly what was posted. **f. Two-up footer:** ink "RUN IT AGAIN" / ground "SHELF".

07 • Feed

FRIENDS / GLOBAL switch with a right-aligned population note. Posts are full-width rows, not cards: handle at 13px/800 (taps the profile), an accent kicker naming the event (PERSONAL BEST, WON A DUEL, CALLED YOU OUT, WORLD RECORD), timestamp right. Below, a 19px headline and meta line on the left, the post's number at 27px with its unit on the right. Footer row: ▲ reaction count, comment count, and an outlined CTA that is always a verb aimed at the reader ("BEAT IT", "RACE IRA", "TRY ORBIT") and always launches a run. An optional top comment hangs below on a 2px neutral left rule. Posts that involve the reader directly sit on accent tint.

Global tab carries world records, today's hardest variant, drop adoption and live cup status — same row anatomy, population in place of a handle.

08 • Duels

Header: kicker "ASYNC · NOBODY WAITS" over a 32px count — "5 GHOSTS WAITING". Then one row per duel: 44px ink initial block, "RAVI · STACK" at 15px, the stake line ("50 blocks — 3 clear of your best"), an expiry line in accent when under twelve hours, and an accent "RACE" button. Rows expiring soon sit on accent tint. Below the list, a full-bleed accent banner for the live tournament ("SATURDAY CUP · ROUND 2 / YOU'RE IN THE LAST EIGHT →") and a one-line link to the inbox.

Cold start: seed with a named, skill-matched global rival labelled "GLOBAL RIVAL" and no expiry. Never fabricate friends.

09 • Wallet & shop

Kicker "ONE WALLET · ALL 40 GAMES", balance at 54px with "coins" beside it, and a line naming how many games it was earned across. "SPEND IT" lists power-ups — EXTRA RETRY 50, MID-RUN REVIVE ×3 180, GHOST SCOUT 120, STREAK FREEZE 300 — each a name, a one-line description and a bordered price button (the cheapest is accent-filled). "TOP UP" is a 3-up pack grid in local currency with the middle pack accent-filled. Closing line, verbatim: "Coins buy continues and retries. They never buy access to a game — all 40 are free."

10 • Your profile

88px accent initial block beside handle (24px), level/rank/join line, and streak line in accent. Then a 2-up stat grid — RUNS TODAY, SESSIONS (with average length), GAMES PLAYED (n / 40), DUELS WON with win rate, COINS EARNED lifetime, BEST STREAK. Then the season ladder: rank name at 30px, progress note right-aligned, and a 10px accent progress bar with a reset countdown in the section label. Then the squad block — weekly co-op target, progress, and the player's own contribution. Then "YOUR BEST GAMES", four rows of name · friend rank · best.

11 • Friend profile

Ink initial block, handle, mutuals line, and an accent "played STACK 4 minutes ago" presence line. Then the head-to-head band on accent tint: "14 – 19" at 40px with the note "Duels since June. They lead by five." Then "WHERE THEY BEAT YOU" — one row per game with the delta in accent when they lead, both scores stacked right, and a bordered "RACE" button. Footer: ink "CHALLENGE" plus an outlined "FOLLOWING" toggle.

12 • Daily drop

Full accent field: "DAILY VARIANT · DAY 141" with a countdown, 46px variant name, the rule in one sentence, and a white "PLAY THE VARIANT" button. Then a 2-up strip — FRIENDS PLAYED (8 / 23) and NEXT NEW GAME (THU). Then the variant leaderboard with the player's row tinted. Closing line explains the cadence: variants rotate daily, a new game lands every Thursday and stays for good.

13 • Tournament bracket

Kicker naming cup, game and entrant count; "LAST EIGHT" at 30px; an accent urgency line with the closing time. Then three columns — LAST 8 / SEMIS / FINAL — of stacked pairings; each pairing is a 1px-bordered box of two rows (name, score). The player's row is accent-filled; a decided winner's row is neutral-200; undecided scores are em dashes. Footer: accent "PLAY YOUR QUARTER-FINAL" and a line naming the score to beat.

14 • Inbox

"INBOX" at 26px with a one-line summary of what is inside. Rows carry a 6px vertical mark (accent when actionable, neutral otherwise), a kicker naming the type (CHALLENGE, CUP, QUEST DONE, SQUAD, FRIEND REQUEST, DROP), the message at 13px/600, a timestamp, and a single-verb button — RACE, PLAY, CLAIM, HELP, ACCEPT, OPEN. Actionable rows sit on accent tint. Every game-related action launches the run directly; none of them open another screen first.

6. Component inventory

Eleven patterns cover all fourteen screens. Build these, not per-screen variants.

COMPONENT	ANATOMY	STATES
Poster band	Accent field, kicker + countdown, 42–46px name, rule line, inverse button	Variant (daily) / new game (weekly)
Editorial game row	54px code block, name, meta, reason, score column	Rival ahead · never played · neutral
Shelf tile	Name, family, best-or-never	Played · never played · pinned
Index row	Number, name, pin flag, sub-line, friends, best	Pinned (tinted) · unpinned · pressed (long-press)
Stat strip	2 or 3 equal cells, kicker over figure	—
Ghost band	Accent tint, rival + score, context line, RACE	Rival ahead · player ahead · global rival
Leaderboard row	Position, name, score	Self (tinted) · friend · global
Feed post	Handle, kicker, time, headline, meta, figure, action row, optional comment	Involves you (tinted) · neutral · global moment
Queue ring	46px conic ring, digit, next-game block, cancel affordance	Counting · cancelled
Notification row	Mark, kicker, text, time, verb button	Actionable (tinted) · informational
Toast strip	Full-width ink bar, 11px tracked label	Pin · unpin · refusal

7. The three playable games

Geometry is ink on ground; the player's active piece is the only accent element. No sprites, no particles, no illustration. Tuning values below are from the reference build and are a starting point, not scripture — but ship the *shape* of the curve.

GAME	INPUT	LOOP AND TUNING
STACK TAP · blocks	Tap anywhere	A 140px block slides across the field at 190px/s, +9px/s per placement to a 430px/s cap. Tap drops it: overlap with the block below is kept, the overhang is lost, and the next block inherits the new width. Alignment inside 6px is a perfect — width is preserved and the placement scores 2 instead of 1. Run ends when overlap falls to 6px or less. Show the last eleven blocks, alternating ink and neutral-800.
REFLEX TIMING · ms	Tap anywhere	A needle sweeps a 100-unit track at 52 units/s, +3.4 per round. An accent band sits at a random centre (18–82) and shrinks from 22 units by 0.8 a round, floor 7. A tap inside scores and reports a millisecond figure; outside is a miss. Twenty rounds or three misses ends the run. Show round, misses and last reaction beneath the track.
DODGE SWIPE · metres	Two lane buttons + arrow keys	Three lanes. Distance accrues at 120 units/s with 5% compounding; obstacles spawn every 0.72s tightening to 0.34s, falling at 250px/s + distance/9. The player sits 56px from the bottom; same-lane overlap ends the run. Score is distance × 10.

Shared rules. Every run races a ghost: the rival's recorded score ticks up over roughly sixteen seconds regardless of game, so the header numbers are always in tension. Coins earned are $\max(8, \text{score} \div 1.6)$ — divided by 40 for distance-scored games — doubled during a variant. A run inherits a carried score when it is a paid or free retry.

8. Copy rules

Loud arcade, but never babbling. The voice is a scoreboard with an attitude: short, factual, second person, and specific about numbers.

DO	DON'T
State the number: "RAVI BEAT YOU BY 3"	"A friend is ahead of you!"
One verb on every button: RACE, BEAT IT, CLAIM, HELP	"Let's go!", "Continue", "OK"
Name the rule in one sentence: "Lights cut out every third block."	Tutorial paragraphs or modal walkthroughs
Uppercase for labels and kickers only	Uppercase sentences, or exclamation marks in body copy
Say what the app did: "Posted to the feed."	Silent side-effects the player has to infer

9. States to build

STATE	TREATMENT
No friends yet	Duels seeds a named global rival; the feed opens on GLOBAL; reason lines fall back to "n FRIENDS PLAY THIS · YOU NEVER HAVE" using global counts. Never an empty screen and never a fake friend.
Never-played game	Best renders as an em dash in neutral-400; the sub-line reads "NEVER PLAYED" in accent-700.
Shelf full	Pin refused with the ink toast; no dialog, no forced swap.
Retries exhausted	Near-miss button switches to "RETRY · 50 COINS"; if the balance is short, the button routes to Shop rather than failing silently.
Queue cancelled	Row collapses to one line plus an outlined play button — the suggestion survives the cancel.
Variant expired	Poster band swaps to the weekly drop; no empty slot, no countdown at zero.
Offline	Runs are playable and queued locally; scores, feed posts and duel results sync on reconnect. A single tracked label in the top bar reads "OFFLINE · SYNCING".

10. Instrumentation

The thesis is a DAU/MAU thesis, so the UI has to be measurable at the seams where it claims to earn stickiness.

EVENT	PROPERTIES
run_started	game, source (shelf · ranked row · queue · duel · inbox · feed · poster), carried_score, variant
run_ended	game, score, improved, beat_ghost, coins, duration
queue_shown / queue_accepted / queue_cancelled	next_game, reason_type, seconds_elapsed
retry_taken	free, delta_to_best, coins_spent
pin_changed	game, pinned, refused
social_action	type (react · comment · challenge · race_from_feed), target_game
session	runs, distinct_games, ended_by (queue_cancel · app_background · navigation)

The two numbers that decide whether this product works: **distinct games per session** (does volume actually get consumed?) and **queue acceptance rate** (does auto-queue extend sessions or annoy people into leaving?).

11. Still open

Decide these before or during build; each one changes screens.

1. **Mid-run revive.** The shop sells it; no mid-run UI has been designed. Either build a three-second revive overlay or cut the item.
2. **Comment composition.** Posts show comment counts and a top comment, but there is no compose or thread screen yet.
3. **Search behaviour.** The Browse field is presentational. Decide fuzzy-name only versus name + family + mechanic keywords.
4. **Squad screens.** Squads appear on the profile and in the inbox with no home of their own.
5. **Notification permission moment.** Async duels depend on push; the ask has no designed placement (recommended: after the first duel win, never on first open).
6. **Dark mode.** The system is a light ground by design. If the audience plays at night, this needs a deliberate inversion, not an automatic one.

Companion documents: *Game Arcade* — *build brief* (implementation and APK packaging) and the interactive reference build.